



Time: 5 minutes

Outcome: Distinguish nominal and real GDP and calculate the GDP deflator.

Difficulty: Intermediate

## Real versus Nominal GDP

### THE CORE IDEA

Nominal GDP values current production at current prices, while real GDP values it at base-year prices. Multiply current quantities by the designated base-year prices and add across final goods. GDP is a strong measure of market production for its purpose, but it omits some dimensions of welfare. Compare nominal and real GDP to separate price changes from changes in production. Real-GDP growth equals the change in real GDP divided by the initial real GDP.

### HOW TO RECOGNIZE IT

- Nominal GDP uses current-year prices and current-year quantities.
- Real GDP uses base-year prices with current-year quantities.
- A change in nominal GDP can reflect prices, quantities, or both; a change in real GDP reflects quantities.
- Use the GDP deflator to compare nominal GDP with real GDP for the same year.

### WATCH OUT

Do not use the new level rather than the initial real-GDP level as the percentage-growth denominator. Real-GDP growth equals the change in real GDP divided by the initial real GDP.

### WORKED EXAMPLE: SEPARATING PRICE AND OUTPUT CHANGES

#### One-good economy: price and output

|              | Quantity | Price | Nominal GDP | Real GDP |
|--------------|----------|-------|-------------|----------|
| Base year    | 100      | \$5   | \$500       | \$500    |
| Current year | 110      | \$6   | \$660       | \$550    |

An economy produces 100 units in the base year at \$5 each and 110 units this year at \$6 each. Nominal GDP this year is  $110 \times \$6 = \$660$ . Real GDP at base-year prices is  $110 \times \$5 = \$550$ . The GDP deflator is  $(\$660 / \$550) \times 100 = 120$ , separating the price increase from the output increase.

### CHECK YOURSELF

Is real GDP calculated by simply subtracting the inflation rate from nominal GDP?